Part 1: Problem Requirements and Conceptual Model

Project Overview: College Event Planning & Budget Management Network

Student organizations submit events for administrator approval. Once approved, students can browse and RSVP to events. All users authenticate with verified school emails.

Selected Functionality: Real-Time Event RSVP Management & Waitlist System

Why Redis? Our event platform faces a concurrency problem during popular event registration. When high-demand events open for RSVPs, multiple students may simultaneously attempt to register, causing race conditions where the event becomes overbooked beyond capacity. Redis solves this through atomic operations that guarantee thread-safe capacity enforcement under high concurrent load.

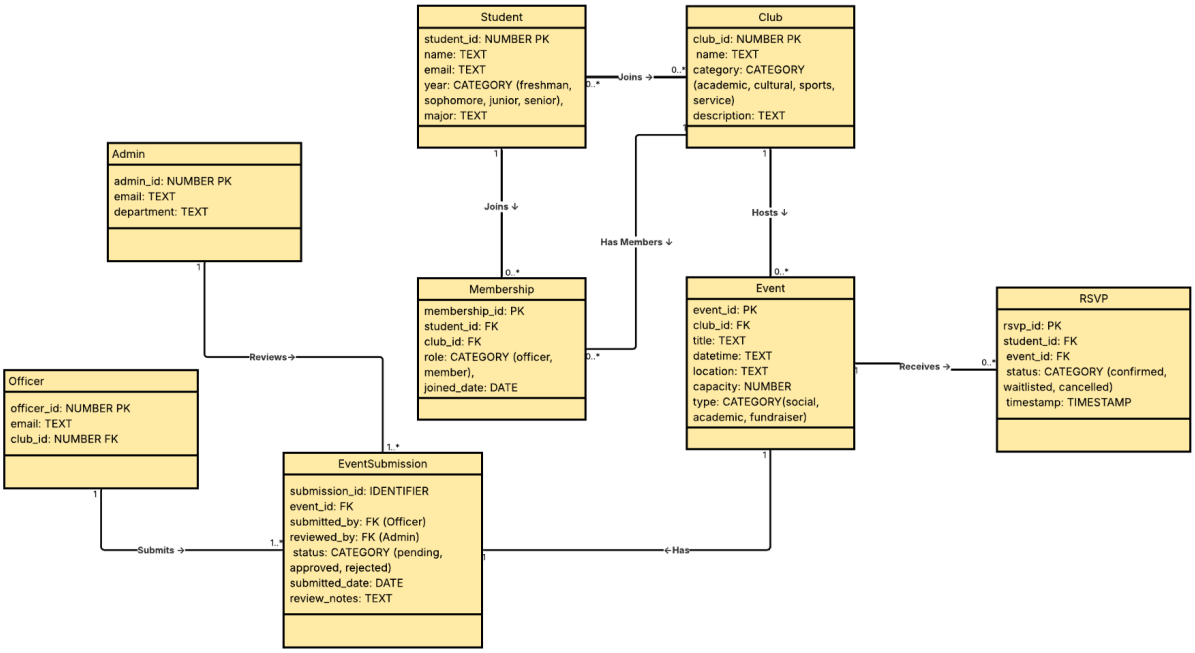
What Gets Stored in Redis:

- Real-time RSVP capacity tracking (current confirmed count, waitlist count, capacity limit)
- Chronologically ordered waitlist queue with student positions
- Confirmed attendee set for fast duplicate checking
- Automatic waitlist promotion when attendees cancel

Business Rules:

- Events enforce maximum capacity on first-come, first-served basis
- RSVP operations are atomic to prevent overbooking
- Waitlist ordered by timestamp
- Cancellations trigger automatic promotion from waitlist

Integration: Redis serves as a high-performance caching layer for real-time state. The relational database maintains permanent RSVP records and event details.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email: TEXT
year: CATEGORY (freshman, sophomore, junior, senior),
major: TEXT
Club
club_id: NUMBER PK
name: TEXT
category: CATEGORY (academic, cultural, sports, service)
description: TEXT
Joins
0..*
0..*
Admin
admin_id: NUMBER PK
email: TEXT
department: TEXT
1
Joins
0..*
Has Members
0..*
1
Hosts
0..*
Membership
membership_id: PK
student_id: FK
club_id: FK
role: CATEGORY (officer, member),
joined_date: DATE
Event
event_id: PK
club_id: FK
title: TEXT
datetime: TEXT
location: TEXT
capacity: NUMBER
type: CATEGORY(social, academic, fundraiser)
RSVP
rsvp_id: PK
student_id: FK
event_id: FK
status: CATEGORY (confirmed, waitlisted, cancelled)
timestamp: TIMESTAMP
Receives
0..*
1
Reviews
1..*
Officer
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email: TEXT
club_id: NUMBER FK
1
EventSubmission
submission_id: IDENTIFIER
event_id: FK
submitted_by: FK (Officer)
reviewed_by: FK (Admin)
status: CATEGORY (pending, approved, rejected)
submitted_date: DATE
review_notes: TEXT
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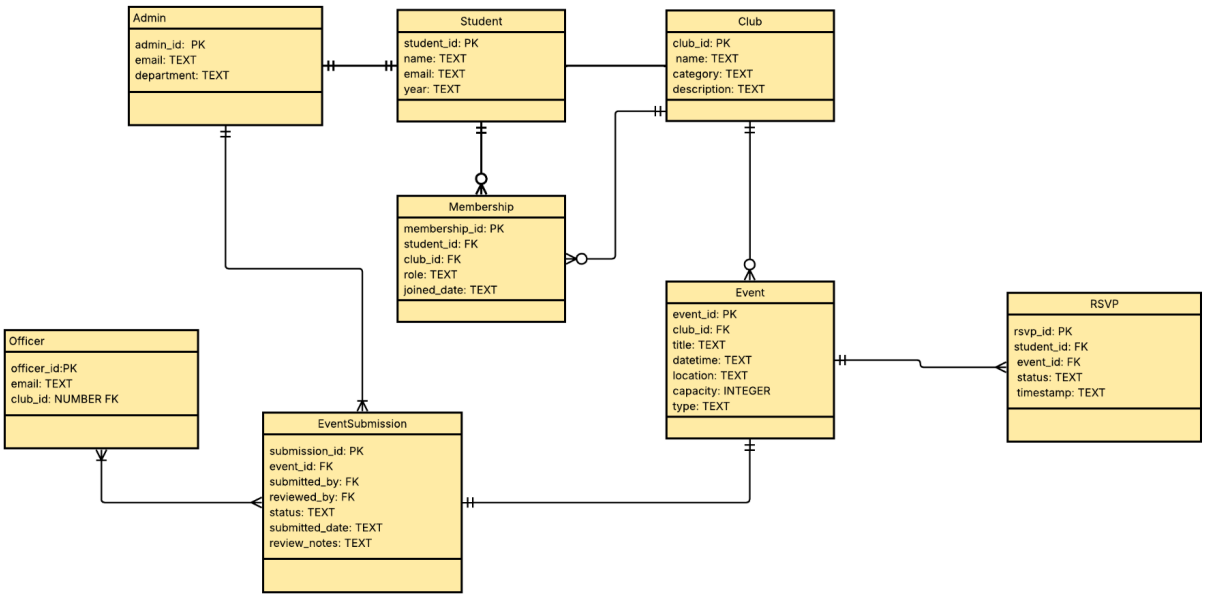
Submits
1..*
1
Has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email: TEXT
club_id: NUMBER FK
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submitted_date: TEXT
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